Transferner: Risk-Aware, On-Device Identification of Edible Plants and Their Toxic Look-Alikes in the Southeastern United States

Jacob Bagley
University of South Carolina
jccbagley@protonmail.com

EmmaLee Peyton
University of South Carolina
emmaleejp@gmail.com

I. Problem Statement and Significance

People with limited botanical knowledge may rely on phone apps to decide whether a wild plant is safe to forage. In a clinical-toxicology evaluation of four popular apps, species-level accuracy was 58%, and five of eleven potentially toxic species were identified as an edible species by at least one app [1]. Standard classifiers treat all errors equally, yet calling poison hemlock wild carrot is far worse than the reverse. We cover ∼1,000 regional species while explicitly targeting *toxic look-alikes* and the asymmetric cost of errors.

We will build **Transferner**, an Android app for South Carolina and neighboring states (North Carolina, Georgia, and Tennessee). A lightweight MobileNetV4 model, designed to be mostly Pareto-optimal across mobile CPUs, DSPs, GPUs, and the Pixel EdgeTPU [2], runs live on the camera feed. When the user captures a still image, BioCLIP 2.5 Huge, the ViT-H/14 successor to BioCLIP 2 [3], identifies the plant. The app returns a species prediction, contact and ingestion toxicity flags, and an explicit "not sure—possible toxic look-alike" response instead of a confident wrong answer. Our research question: *Which adaptation and compression techniques (LoRA, partial INT8 quantization, etc.) preserve or improve toxic/edible discrimination under a phone's latency and memory budget?*

II. Data

Sources. (i) Pl@ntNet observations from GBIF [4] (records CC BY 4.0; each photo carries its own license), filtered by GADM state; the four states hold ∼13.9K photographed observations (Oct. 2026); and (ii) TreeOfLife-200M [3], which combines GBIF, EOL, BIOSCAN-5M, and FathomNet into 214M images of 952,257 taxa. It ships metadata and URLs, so we download only our target species.

Scope. About 1,000 regional species. Most have no meaningful visual overlap with a toxic species and are trained as ordinary classes; a subset forms toxic/edible look-alike groups that receive targeted data and evaluation, e.g., poison hemlock (*Conium maculatum*) vs. wild carrot (*Daucus carota*); water hemlock (*Cicuta maculata*) vs. wild carrot and elderberry (*Sambucus canadensis*); Canada moonseed (*Menispermum canadense*) and pokeweed (*Phytolacca americana*) fruit vs. wild grapes incl. muscadine (*Vitis* spp.); and, for contact toxicity, poison ivy (*Toxicodendron radicans*) vs. blackberry/raspberry (*Rubus* spp.). Each pair is documented as a confusion by extension or poison-center sources. Toxicity labels (humans only) are *independent* binary contact and ingestion flags derived from the NC State Extension Plant Toolbox: contact from its "Causes Contact Dermatitis" field, ingestion from its poison severity and symptoms.



Fig. 1. Young plants of a look-alike pair: wild carrot (*D. carota*, left; root edible when young, hairy stem) vs. poison hemlock (*C. maculatum*, right; toxic, hairless stem with purple blotching). Photo © 2015 H. Holly, used with credit [5].

Size and splits. About 200 images per toxic and look-alike species, drawn from both TreeOfLife-200M and Pl@ntNet; 70/15/15 train/val/test split by *observation*, so that photos of the same plant never appear in two splits; test observations come only from the leakage-free pool below.

Leakage control. BioCLIP 2/2.5 were pretrained on TreeOfLife-200M, so test images drawn from it would inflate scores. Our test set will use Pl@ntNet/GBIF records *not* in the TreeOfLife-200M catalog (matched by source ID, plus perceptual-hash de-duplication) or records newer than its snapshot.

Known issues. Long-tailed class counts, label noise in citizen-science data, and appearance that changes with growth stage (seedling, flower, fruit).

III. Proposed Method

Two-stage pipeline. *Stage 1 (live):* MobileNetV4 [2] is fine-tuned from ImageNet weights (timm) on our ∼1,000-way

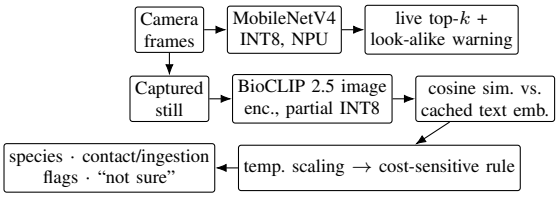


Fig. 2. On-device pipeline. All inference runs on the phone; text embeddings for the ∼1,000 labels are computed once offline.

regional label set. Stage 1 runs per frame to give a live top-k guess and to warn early when a look-alike group is in view. *Stage 2 (still image):* BioCLIP 2.5 Huge (CLIP-style ViT-H/14, loaded with OpenCLIP) classifies the still image. Text embeddings for our label set are computed offline, so only the roughly 632M-parameter image encoder runs at inference. At 632M parameters its weights alone occupy ∼1.26 GB in FP16 versus ∼0.63 GB in INT8, which is why quantization is required, not optional, for on-device use. Fig. 2 summarizes the pipeline. **All app inference runs on the edge device**; servers are used only for training and lab testing.

Techniques we will compare.

- **Zero-shot baselines:** BioCLIP, BioCLIP 2, and BioCLIP 2.5.
- **LoRA** [6] on the BioCLIP 2.5 image encoder (via Hugging Face PEFT), compared against a linear probe on frozen features; MobileNetV4 is fully fine-tuned.
- **Look-alike-focused training:** class-balanced loss for the long tail.
- **Partial INT8 quantization** [7]: post-training and quantization-aware INT8. Layers whose quantization error most raises toxic-class confusion stay at FP16.
- **Risk-aware decision:** temperature scaling [8] gives calibrated $p_i = \mathrm{softmax}(z/T)_i$. With cost C_{ij} for predicting j when the truth is i (large when i is toxic and j is not), the app outputs $\hat{y} = \arg\min_j \sum_i C_{ij}\, p_i$, or "not sure" when that minimum expected cost exceeds an abstention cost c_a. C and c_a are tuned on validation data.

Implementation. Training uses PyTorch on an AMD Radeon RX 7900 XTX (24 GB, gfx1100), which ROCm officially supports on Linux. Models are exported with LiteRT Torch to LiteRT and run on a Google Pixel phone, using the NPU and CPU in tandem. LiteRT's Google Tensor NPU path is in beta and supports ahead-of-time compilation only, so CPU/GPU execution is our fallback. Code: https://github.com/CyberneticMedic/Transferner. **Roles:** J. Bagley—data, training, quantization, on-device benchmarking; E. Peyton—app front end/UX, safety-message design, report.

IV. Evaluation and Expected Results

Headline metric—dangerous miss rate (DMR): with $\mathcal{T}$ the toxic classes and $\mathcal{D}_\mathcal{T}$ the test images whose true class is toxic, $\mathrm{DMR} = |\{x \in \mathcal{D}_\mathcal{T} : \hat{y}(x) \notin \mathcal{T} \cup \{\text{not sure}\}\}| \,/\, |\mathcal{D}_\mathcal{T}|$, i.e., toxic plants the app confidently calls non-toxic. It is computed separately with $\mathcal{T}$ as the contact-toxic and the ingestion-toxic set. We report it overall and per look-alike pair. **Other metrics:** top-1/top-5 accuracy, macro-F1, expected calibration error (ECE, overall and on toxic classes), coverage at fixed DMR, on-device latency (NPU vs. CPU), model size, and peak memory. Lab tests run on held-out data; deployment tests are *field-based*, photographing plants in situ in the target region. We give 95% bootstrap confidence intervals and use McNemar's test to compare paired model variants. **Expected figures:** look-alike confusion matrices; reliability diagrams before and after calibration; risk–coverage curves; and an accuracy/DMR vs. latency Pareto plot across FP32, partial-INT8, and full-INT8 models. **Hypotheses:** (H1) LoRA-adapted BioCLIP 2.5 lowers DMR versus zero-shot. (H2) Full INT8 hurts look-alike pairs more than overall accuracy; partial INT8 recovers most of the loss. (H3) Calibration plus abstention cuts DMR at a modest coverage cost.

Risks and fallbacks. (i) Google Tensor NPU support is beta: fall back to GPU/CPU delegates and report per-unit latency. (ii) Regional Pl@ntNet data is thin (∼14 observations/species on average): supplement from TreeOfLife-200M and report per-source accuracy. (iii) BioCLIP 2.5 too slow on-device even at INT8: distill it into MobileNetV4, or use BioCLIP 2 (ViT-L/14) as the still-image model.

V. Timeline (Proposed)

Dates	Milestone
Oct 19–25	Project coordination and planning; repository setup; species list and toxicity labels; data scripts
Oct 26–Nov 1	Dataset built and split; zero-shot BioCLIP and MNv4 baselines; app camera + LiteRT skeleton
Nov 2–15	LoRA on BioCLIP 2.5; class-balanced MNv4; calibration, cost rule, abstention; app result/warning UI
Nov 16–22	INT8 (PTQ/QAT, partial); LiteRT export; Pixel NPU/CPU benchmarks
Nov 23–29	Full evaluation, failure analysis (Thanksgiving buffer)
Nov 30–Dec 10	Report, demo video, slides; rehearse; present Dec 10; submit zip

References

[1] K. Long *et al.*, "Plant identification applications do not reliably identify toxic and edible plants in the American Midwest," *Clin. Toxicol.*, vol. 61, no. 7, pp. 524–528, 2023.

[2] D. Qin *et al.*, "MobileNetV4: Universal models for the mobile ecosystem," in *Computer Vision – ECCV 2024*, ser. Lecture Notes in Computer Science, vol. 15098. Cham, Switzerland: Springer, 2024, pp. 78–96.

[3] J. Gu *et al.*, "BioCLIP 2: Emergent properties from scaling hierarchical contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 102 778–102 811.

[4] A. Affouard *et al.*, "Pl@ntNet observations," Occurrence dataset, version 1.9, GBIF, 2025, accessed: Oct. 18, 2026. [Online]. Available: https://doi.org/10.15468/gtebaa

[5] H. Holly, "Wild carrot vs poison hemlock – a deep dive," The Northwest Forager (blog), Jan. 2026, accessed: Oct. 18, 2026. [Online]. Available: https://thenorthwestforager.com/2026/01/02/wild-carrot-vs-poison-hemlock/

[6] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[7] M. Nagel, M. Fournarakis, R. A. Amjad, Y. Bondarenko, M. van Baalen, and T. Blankevoort, "A white paper on neural network quantization," arXiv:2106.08295, 2021.

[8] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, vol. 70, 2017, pp. 1321–1330.